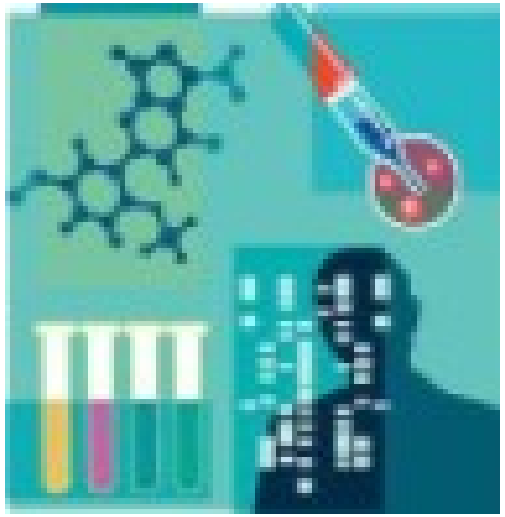
Techniques in Biotechnology



Review of:

Gene Therapy

Cell Replacement Therapy

Tissue Engineering

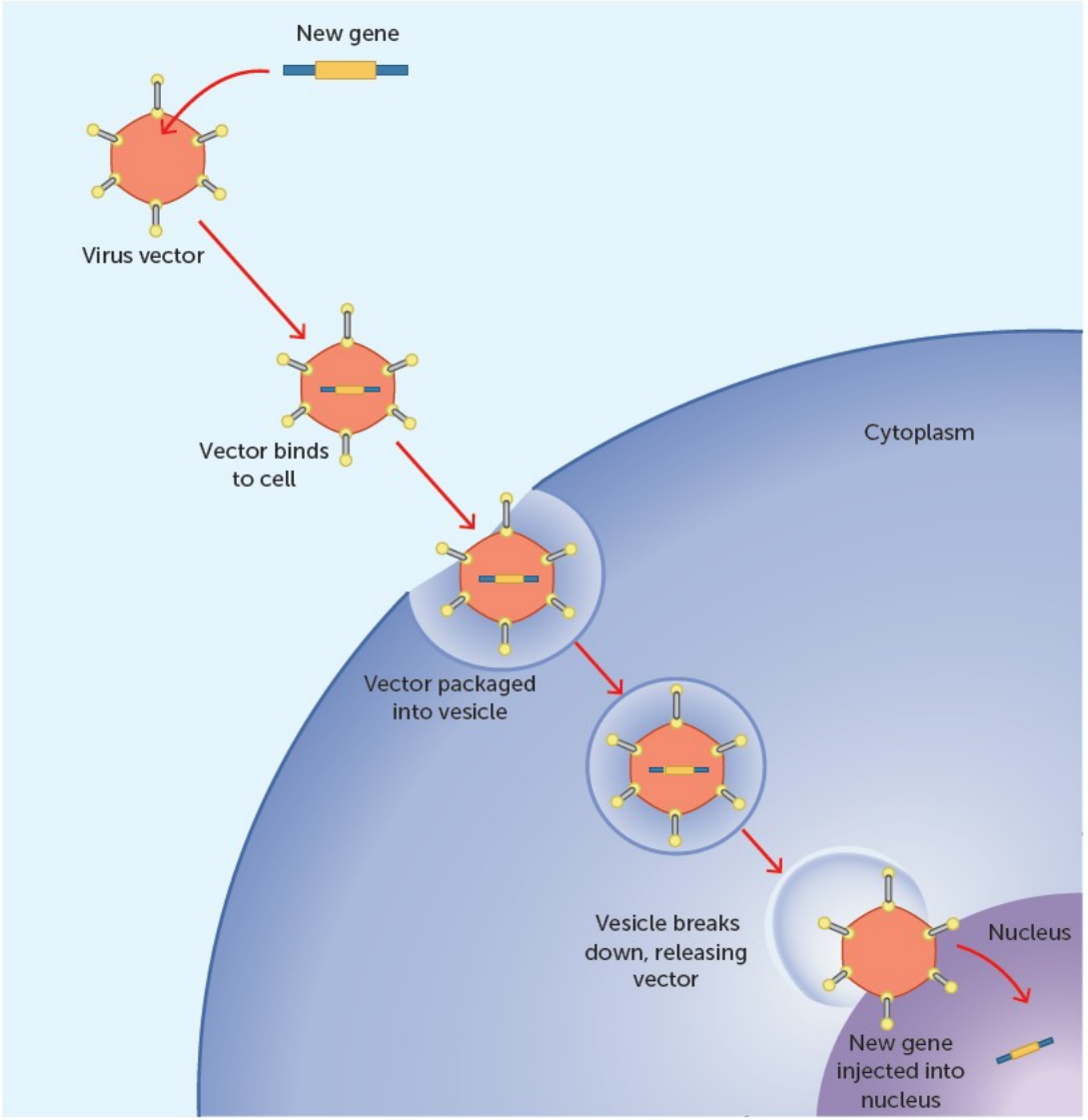
Table 13.3 Summary of techniques for identifying genes and alleles

Technique	Features
DNA sequencing	Determines the precise order of nucleotides in a sample of DNA Enables sequences to be compared easily
Profiling	Determines an individual's DNA profile or fingerprint as a distinct series of bands Enables the detection of many genetically inherited diseases
Polymerase chain reaction	Segments of DNA are multiplied through a series of repeated cycles using an enzyme called DNA polymerase Genes of interest can easily be amplified and then sequenced to detect the mutations

Gene Therapy

- Aims to treat or cure disease by identifying faulty genes and inserting healthy ones.
- Still in research phase.
- Concept is that a vector (eg a virus)can be used to deliver the desired DNA into the cell, to be incorporated by the nucleus and produce desired protein.
- Areas of possibility:
  - Replacing mutated gene with healthy copy*
    - Eg **Huntington's disease**, Cystic Fibrosis, Type I Diabetes
  - Fixing or inactivating mutant genes*
    - Eg **Huntington's disease**
  - Inserting a new gene that will fight the disease
  - Making the immune system recognise diseased cells.
    - Eg therapy for some cancers

*These ones could be used for Huntington's Disease



Gene Therapy: Cystic Fibrosis

- Cystic Fibrosis is relatively common
- Recessive condition, due to faulty gene:
  - Cystic Fibrosis Transmembrane Regulator (CFTR)
    - Discovered in 1989
  - Causes inability of body to process certain secretions
    - Buildup of mucus in lungs, and digestive tract
    - Can significantly shorten life
- First experimental gene therapy given in 1993
  - Used the common cold virus as vector to carry the normal gene into lungs
  - Mainly assessed safety of procedure
  - Not yet used as therapy, still in research phase.

Cell Replacement Therapy

- Uses stem cells
  - Undifferentiated
  - Can continue repeated divisions over long periods of time
  - Can differentiate into other cell types in the right conditions
  - Often sourced from embryos
- Stem cell replacement therapy
  - Especially for neurological diseases eg:
    - Parkinson’s
    - Alzheimer’s
  - Some success in pilot studies so far.
  - Controversial due to use of embryonic tissue

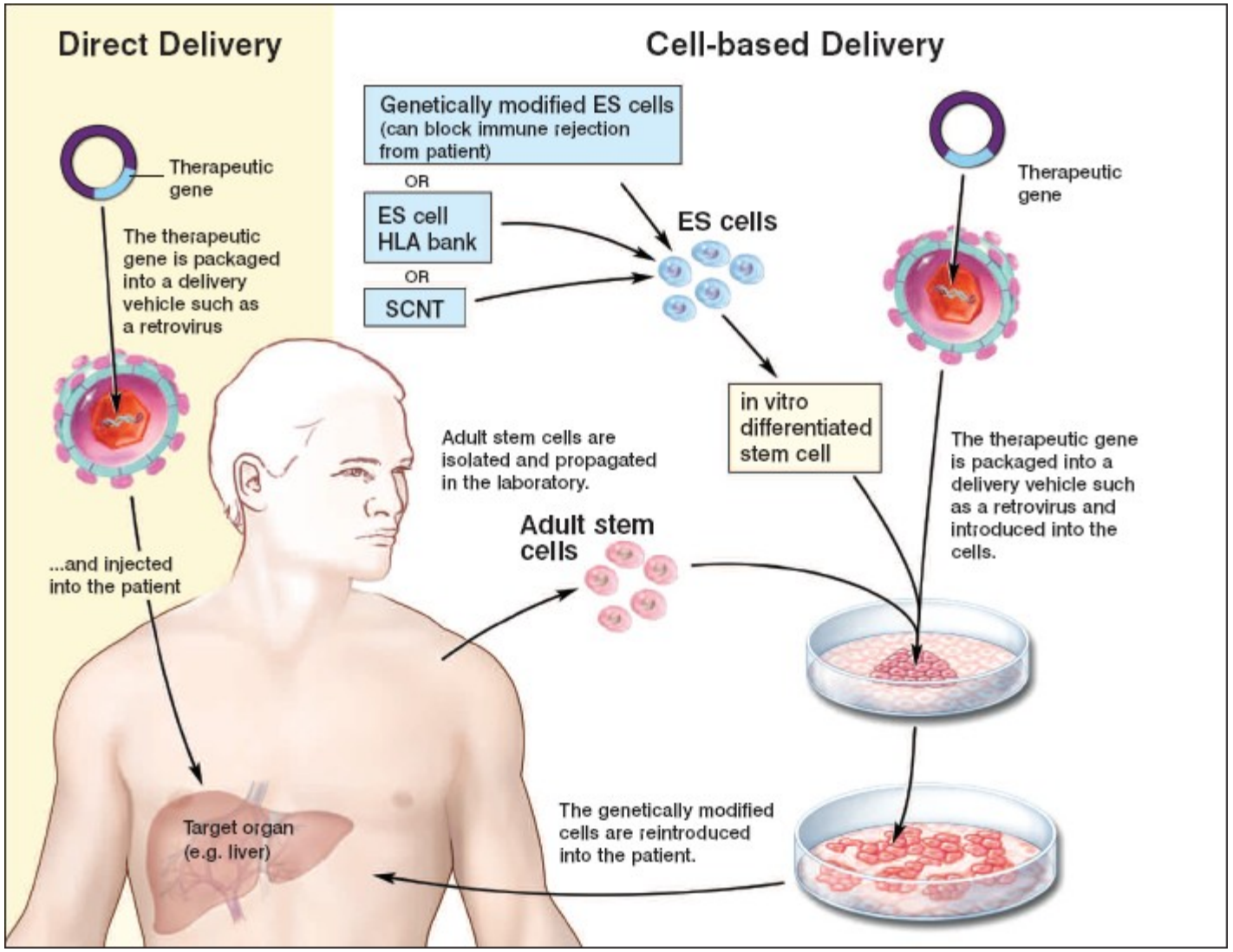
Cell Replacement Therapy

- Could be used in future for neurodegenerative disorders such as Alzheimer's and Parkinson's.

- Replacement of dying neurons with healthy ones derived from stem cells
  - Can grow and differentiate to replace damaged tissue
  - Controversy surrounding use of human embryonic stem cells
  - Other sources of stem cells are being researched

- Technique not yet refined – is a potential future treatment.

How Cell Replacement therapy might work once it is through the experimental stage:



This example shows the liver as the target organ, but the idea is the same for diseases such as Alzheimer's and Parkinson's where a part of the brain is the target.

Tissue Engineering

- Also uses stem cells to:
  - Restore healthy tissues or organs
  - Eliminate the need for transplants and implants

- Cells grown over natural or synthetic scaffold to make new organ or tissue
  - Scaffolds need to have
    - High pore sizes so cells can grow and nutrients can diffuse
    - Be degradable so they break down as the cells grow
  - Stem cells must be cultured
    - Seeded onto scaffold
  - Could be used to develop tissues such as bone, skin, cartilage, adipose tissue.